

Epileptic Seizures as Aftershock Sequences: Self-Excitation Improves Seizure Forecasting Beyond Cycle Models

Dat Nguyen Independent Researcher Dtnghuyen09@gmail.com

Abstract

Seizure forecasting methods based on circadian and multidien cycles have emerged as a promising approach for predicting seizure likelihood from seizure diaries alone. However, these cycle-based models do not account for a second, potentially independent temporal structure: the tendency of seizures to trigger subsequent seizures, analogous to earthquake aftershocks. Here we introduce the Epidemic-Type Aftershock Sequence (ETAS) model from statistical seismology as a complementary seizure forecasting framework and test it head-to-head against cycle-based forecasting in a pseudo-prospective evaluation. We analyzed 694 seizures from 28 patients across two public datasets: the CHB-MIT Scalp EEG Database (24 patients, 198 seizures) and the Melbourne NeuroVista Seizure Prediction Trial (4 patients, 496 seizures, 216–663 days). Inter-seizure intervals rejected the Poisson model in both datasets ($p < 10^{-5}$ and $p < 10^{-87}$) and followed an Omori-type power-law decay with exponent $p \approx 1.0$ ($R^2 = 0.93$ – 0.97). Post-seizure rate elevation was 35.6-fold in the first 30 minutes, decaying to baseline by ~ 48 hours. In pseudo-prospective forecasting on NeuroVista patients, ETAS (mean AUC = 0.634) outperformed cycle-based forecasting (mean AUC = 0.596), and a combined model incorporating both self-excitation and circadian cycles achieved the highest performance (mean AUC = 0.721), winning for every patient tested. The estimated branching ratios of 0.44 and 0.75 indicate that 44–75% of seizures are temporally clustered in a manner consistent with self-triggering. These results demonstrate that self-excitation and cycles capture independent information about seizure timing, and that combining them improves forecasting performance beyond either approach alone. The ETAS framework requires only seizure timestamps, making it immediately deployable in smartphone seizure diary applications.

Keywords: epilepsy, seizure forecasting, self-exciting point process, ETAS, Hawkes process, Omori law, branching ratio, seizure clustering, seizure cycles

1. Introduction

Epileptic seizures are not random events. Recent advances in long-term seizure monitoring have revealed that seizure timing is modulated by circadian (24-hour) and multidien (multi-day) cycles detectable from seizure diaries, interictal epileptiform activity, and wearable biosignals (Károly et al., 2021; Baud et al., 2018). These discoveries have enabled cycle-based seizure forecasting algorithms that represent the current state of the art for diary-based prediction (Károly et al., 2020; Nasser et al., 2025).

However, cycle-based models capture only periodic modulation of background seizure risk. They do not model a second, clinically recognized phenomenon: seizure clustering, in which one seizure appears to increase the probability of subsequent seizures over a timescale of minutes to hours. Seizure clustering affects 20–70% of patients with epilepsy and is associated with increased risk of status epilepticus and sudden unexpected death in epilepsy (SUDEP) (Haut et al., 2006; Cedarbaum et al., 2021).

Osorio et al. (2010) demonstrated a striking statistical analogy between seizure clustering and earthquake aftershock sequences, showing that seizures obey the same Omori power-law decay, Gutenberg-Richter size distribution, and inter-event interval distribution as earthquakes. This analogy suggested that seizure forecasting could benefit from the self-exciting point process models used in statistical seismology.

In seismology, the Epidemic-Type Aftershock Sequence (ETAS) model (Ogata, 1988) is the standard tool for earthquake forecasting. ETAS decomposes seismicity into a background rate of independent events and a self-excitation component in which each event increases the probability of subsequent events, with the enhancement decaying as a power law (Omori's law). The branching ratio — the expected number of triggered events per independent event — quantifies the degree of self-excitation.

Despite the established statistical analogy, no study has applied ETAS as a calibrated seizure forecasting tool, estimated patient-specific branching ratios from clinical data, or compared ETAS forecasting performance against the current cycle-based approach. The seizure forecasting field has proceeded from the statistical analogy (Osorio et al., 2010) directly to EEG-based machine learning (Truong et al., 2018) and cycle detection (Karoly et al., 2020), without exploring physics-based point process models.

Here we address this gap. We fit the ETAS model to seizure timestamps from two public datasets, demonstrate that self-excitation provides forecasting value independent of cyclic modulation, and show that a combined model incorporating both mechanisms outperforms either alone.

2. Methods

2.1 Data

CHB-MIT Scalp EEG Database (Shoeb, 2009): Continuous scalp EEG from 24 pediatric patients with intractable epilepsy undergoing pre-surgical evaluation at Children's Hospital Boston, containing 198 annotated seizures. Monitoring duration ranged from hours to days.

Melbourne NeuroVista Seizure Prediction Trial (Cook et al., 2013): Long-term continuous intracranial EEG from patients with drug-resistant focal epilepsy. We used four patient records available through the forecast-benchmarks repository (RiPL Research), containing 145, 32, 297, and 22 seizures over 663, 596, 432, and 216 days, respectively (496 total seizures).

2.2 Inter-Seizure Interval Analysis

Inter-seizure intervals (ISIs) were tested for departure from the exponential distribution (Poisson process) using the Kolmogorov-Smirnov test. The Weibull distribution was fit to ISIs; a shape parameter below 1 indicates decreasing hazard consistent with clustering. Omori-type power-law decay was assessed by linear regression of $\log(\text{ISI density})$ versus $\log(\text{ISI})$ using logarithmically spaced bins.

2.3 Post-Seizure Rate Elevation

The fraction of ISIs shorter than specified time windows (0.5–48 hours) was compared to the Poisson expectation using binomial tests.

2.4 Forecasting Models

Four models were compared:

Poisson model. Constant intensity $\lambda(t) = N/T$ (seizures per hour). Serves as a naive baseline.

ETAS model. Self-exciting intensity:

$$\lambda(t) = \mu + \sum_i K \cdot (t - t_i + c)^{-p}$$

where μ is the background rate, K is aftershock productivity, c is a regularization constant, p is the Omori decay exponent, and the sum runs over previous seizures $t_i < t$. Parameters estimated by maximum likelihood.

Cycle model. Periodically modulated intensity:

$$\lambda(t) = \mu_0 \cdot \exp[A_1 \cos(2\pi t/24 + \phi_1) + A_2 \cos(2\pi t/T_2 + \phi_2)]$$

capturing circadian (period 24h) and multidien (period T_2) rhythms. Parameters estimated by maximum likelihood with grid search over candidate multidien periods (3–28 days).

Combined model. Cycle-modulated background plus self-excitation:

$$\lambda(t) = \mu_0 \cdot \exp[A_1 \cos(2\pi t/24 + \phi_1)] + \sum_i K \cdot (t - t_i + c)^{-p}$$

combining circadian modulation with ETAS aftershock triggering.

The branching ratio $n = K \cdot c^{(1-p)} / (p - 1)$ for $p > 1$ estimates the fraction of seizures attributable to self-triggering.

2.5 Pseudo-Prospective Evaluation

Models were trained on the first 60% of each patient's seizures. Forecasting ability was evaluated on the remaining 40% by computing the model intensity at actual seizure times versus 2,000 uniformly sampled random times in the test period. Performance was measured by the area under the receiver operating characteristic curve (AUC): the probability that the model assigns

higher intensity to a seizure time than a random time. AUC = 0.5 indicates no forecasting ability; AUC = 1.0 indicates perfect discrimination.

For ETAS and the combined model, the intensity at each test time incorporated all seizures (training and earlier test seizures) occurring before that time, simulating real-time updating.

3. Results

3.1 Seizure Clustering

Inter-seizure intervals rejected the exponential distribution in both datasets: CHB-MIT (KS $p = 2.4 \times 10^{-5}$) and NeuroVista ($p = 5.6 \times 10^{-87}$). The Weibull shape parameter was 0.716 (CHB-MIT) and 0.392 (NeuroVista), both well below 1.0, confirming decreasing hazard consistent with aftershock-like clustering. The combined dataset yielded a Weibull shape of 0.370.

3.2 Omori Law

ISI density followed a power-law decay with Omori exponents of $p = 0.99$ ($R^2 = 0.932$, CHB-MIT) and $p = 0.91$ ($R^2 = 0.937$, NeuroVista). The combined dataset yielded $p = 1.05$ ($R^2 = 0.965$). These values match the universal Omori exponent $p \approx 1.0$ observed in earthquake aftershock sequences.

3.3 Post-Seizure Rate Elevation

In the NeuroVista data, seizure rate was elevated 35.6-fold in the first 30 minutes after a seizure (binomial $p = 3.3 \times 10^{-112}$), decaying to 20.5 $\times$ at 1 hour, 8.7 $\times$ at 4 hours, 2.9 $\times$ at 24 hours, and 1.8 $\times$ at 48 hours. All elevations were highly significant ($p < 10^{-50}$).

3.4 ETAS Model Fits

For the two NeuroVista patients with more than 100 seizures, ETAS strongly outperformed Poisson: $\Delta AIC = +398$ (patient eg1, 145 seizures) and +444 (patient eg3, 297 seizures). Estimated branching ratios were 0.44 (eg1) and 0.75 (eg3), indicating that 44–75% of seizures were temporally clustered in a manner consistent with self-triggering. The ETAS time constant c was 0.08–0.09 hours (~ 5 minutes), indicating triggering operates on a timescale of minutes and decays over hours.

For patients with fewer seizures (32 and 22), ETAS did not outperform Poisson, indicating the 4-parameter model requires approximately 100+ seizures to calibrate effectively.

3.5 Pseudo-Prospective Forecasting

In the pseudo-prospective evaluation on three NeuroVista patients with sufficient data (Table 1), the combined model achieved the highest AUC for every patient tested.

Table 1. Pseudo-prospective forecasting performance (AUC).

Patient	N_train	N_test	ETAS	Cycle	Combined	Poisson	MA(90d)
eg1	87	58	0.686	0.623	0.772	0.500	0.514
eg2	19	13	0.402	0.522	0.575	0.500	0.461
eg3	178	119	0.814	0.644	0.815	0.500	0.470
Mean			0.634	0.596	0.721	0.500	0.482

ETAS alone (mean AUC = 0.634) outperformed the cycle model alone (0.596) by 3.8 AUC points. The combined model (0.721) outperformed both ETAS alone (+8.7 points) and cycles alone (+12.5 points). The 90-day moving average performed below chance (0.482), indicating that simple rate trending is not useful for these patients.

For patient eg3 (297 seizures, branching ratio 0.75), ETAS achieved an AUC of 0.814 — correctly distinguishing seizure from non-seizure times 81% of the time using timestamps alone.

To verify that the combined model's superior out-of-sample performance was not driven by overfitting, we computed the Bayesian Information Criterion (BIC) on the training data, which penalizes model complexity more heavily than AIC. On training data, ETAS alone achieved the lowest BIC for patients eg1 and eg3 ($\Delta\text{BIC} = 163\text{--}268$ points below the cycle model), indicating that self-excitation is the dominant in-sample signal. The combined model matched ETAS in-sample ($\Delta\text{BIC} < 1$ for eg1, +15 for eg3) while achieving superior out-of-sample AUC, suggesting that the cycle component captures generalizable periodic structure that improves forecasting without overfitting.

4. Discussion

4.1 Self-Excitation and Cycles Capture Independent Information

The central finding of this study is that self-excitation and cyclic modulation capture independent aspects of seizure timing, and combining them improves forecasting beyond either mechanism alone. This result has a clear interpretation: cycles determine *when* a patient's baseline risk is high (e.g., certain times of day or phases of multi-day rhythms), while self-excitation determines *whether* the current moment is in the wake of a recent seizure. A patient whose circadian cycle places them in a low-risk period may nonetheless be at elevated risk if they had a seizure 20 minutes ago. Current cycle-only forecasters miss this short-term risk elevation entirely.

4.2 The Branching Ratio as a Clinical Metric

The branching ratio provides a quantity not available in the current clinical framework: the fraction of a patient's seizures that are temporally clustered in a pattern consistent with self-

triggering. For patient eg3 (branching ratio 0.75), three out of four seizures occur in temporal clusters consistent with aftershock-like triggering. This patient might benefit from aggressive early intervention after the first seizure in a cluster — rescue medication, responsive neurostimulation, or caregiver alerting — because interrupting the cascade could reduce the clustered seizure burden.

We note that a high branching ratio indicates temporal clustering consistent with self-excitation but does not prove direct causal triggering. Alternative explanations for short-interval clustering include shared underlying state changes (e.g., sleep stage transitions, medication level fluctuations, or hormonal cycles operating on timescales shorter than those captured by our cycle model). Disentangling statistical clustering from causal triggering would require interventional studies in which post-seizure treatment is randomized.

4.3 Relationship to Prior Work

Osorio et al. (2010) established the earthquake-seizure analogy across five statistical laws. Simkin and Roychowdhury (2010) provided a theoretical explanation via branching process models. The present work advances this line of research by fitting the explicit ETAS model to long-term individual patient data with maximum likelihood estimation, extracting patient-specific branching ratios, and demonstrating — for the first time — that self-excitation provides forecasting value independent of cycle-based approaches in a pseudo-prospective evaluation.

The cycle-based forecasting literature (Karoly et al., 2020; Nasser et al., 2025) has established that circadian and multidiurnal rhythms are useful for diary-based seizure prediction. Our results are complementary, not contradictory: cycles and self-excitation capture different temporal structures, and the combined model benefits from both.

4.4 Limitations

This study has several important limitations. First, the NeuroVista example dataset contains only four patients, three of whom had sufficient seizures for pseudo-prospective evaluation. Validation on larger cohorts — the full NeuroVista trial (15 patients), NeuroPace RNS cohorts (175+ patients), or large seizure diary datasets (768+ users in the Seer app cohort) — is essential before clinical conclusions can be drawn. Second, the CHB-MIT data was collected during pre-surgical monitoring with medication withdrawal, which may artificially increase clustering; however, the NeuroVista data represents chronic ambulatory monitoring and shows even stronger clustering. Third, our cycle model implementation is simplified relative to the state-of-the-art approaches that use wavelet transforms and spectral analysis for cycle detection (Karoly et al., 2020); a more sophisticated cycle model might narrow the performance gap between cycles and ETAS. Fourth, we treated all seizures as equivalent events; incorporating seizure severity or type could improve both models. Fifth, the pseudo-prospective evaluation uses a fixed 60/40 train-test split rather than the expanding-window approach used in some seizure forecasting studies.

4.5 Implications for Seizure Diary Applications

The ETAS model requires only seizure timestamps — the same input already collected by smartphone seizure diary apps such as Seer Medical, Seizure Tracker, and Nile. Implementing ETAS as an additional forecasting layer would require no new hardware or sensors. After each recorded seizure, the app could display a time-varying risk estimate based on the Omori decay of the patient's personal aftershock rate, complementing any existing cycle-based forecast.

5. Conclusion

Self-excitation — the tendency of seizures to trigger subsequent seizures following an Omori power-law decay — captures independent forecasting information not available in cycle-based models. A combined model incorporating both self-excitation and circadian cycles outperformed either approach alone for every patient tested, with the combined model achieving a mean pseudo-prospective AUC of 0.721 compared to 0.634 for ETAS alone and 0.596 for cycles alone. The branching ratio, borrowed from statistical seismology, provides a new interpretable metric quantifying the fraction of seizures attributable to temporal clustering. These results motivate the integration of self-exciting point process models into existing seizure forecasting frameworks.

Data Availability

CHB-MIT data: <https://physionet.org/content/chbmit/1.0.0/>. NeuroVista example data: <https://github.com/RiPLresearch/forecast-benchmarks>. Analysis code will be made available upon publication.

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